Project 1: LED Circuits

Welcome to the first OPS project! Each project applies concepts taught during lecture and will be due by the following lecture. If you need help, reach out on Slack or attend office hours!

Project 1 introduces breadboarding using buttons (input) and LEDs (output). Through this project, we will build a foundation for all future projects by mastering **breadboard prototyping** and **circuit design**.

Objectives

- Introduction to OPS kits
- Understand how to translate a circuit diagram into a breadboard prototype
- Understand how to design a circuit diagram given a list of specifications
- Understand how the potential difference between the nodes of an LED causes it to turn on
- Understand common LED circuits and button configurations

Task Description

Summary:

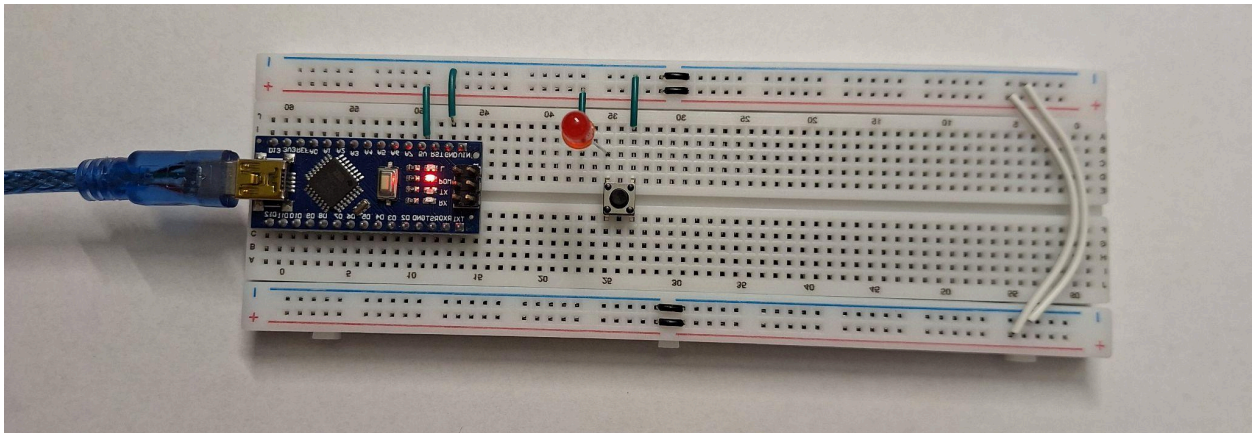
We will create three common circuits on your breadboard that turn on LEDs when buttons are pressed. Use your Arduino Nano to supply power and ground to your breadboard, and use your wire kit to connect the components to form a functional circuit.

What you will need:

- 1x Arduino Nano (and Power Cord)
    - https://www.digikey.com/en/products/detail/raspberry-pi/SC0918/16608263?s=N4IgTCBcDaIM4GMAMBOAjADhAXQL5A&src=raspberrypi
- 3x LED
    - https://www.digikey.com/en/products/detail/qt-brightek-qtb/QBL8IW15C-CW-2897/14664272
- 4x Pushbutton
    - https://www.digikey.com/en/products/detail/cui-devices/TS02-66-55-BK-160-LCR-D/15634374
- 1x Wire Kit
- 5x 1k Ohm Resistor

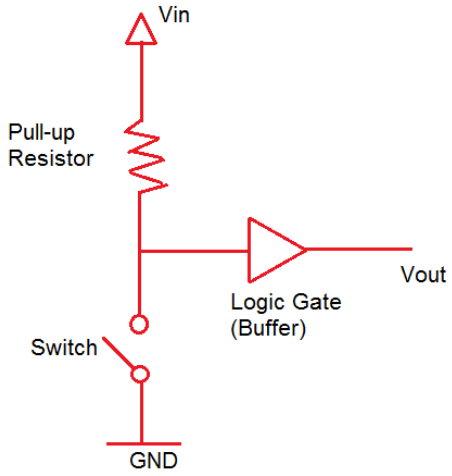
- Optional: You may wish to purchase a pair of needle-nose pliers to manipulate the small wires while breadboarding. Additionally, a multimeter may be a good choice for debugging. These items are not required for OPS, but strongly recommended.

Part 1: Pull-Up & Pull-Down LED Circuits



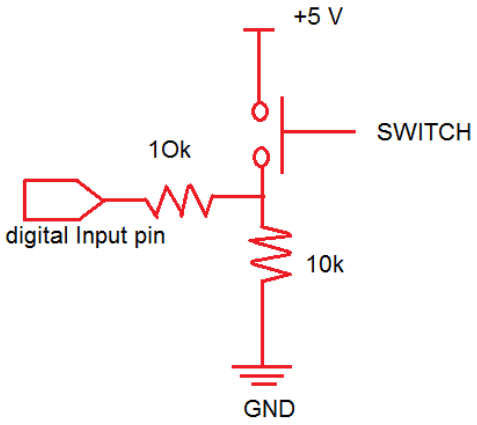
During lecture, we learned that simple button configurations such as this are undesirable because they result in "floating" signals that may produce indeterminate states when the button is not pressed. ***Note: the reason that this is important may not seem obvious yet, but will become extremely important later on when using the Nano's input pins. When reading the below schematics, interpret Vout as the "input" to an LED.**

In part 1 of this project, we will compare two similar LED circuit configurations that solve the floating signal problem by driving the LED even when the pushbutton is not pressed.



Pull-Up Resistor Schematic by Jean Braza

First, construct the pull-up resistor circuit on your breadboard. In this configuration the LED is connected to GND when the pushbutton is pressed, so it turns off. The resistor connects the LED to 5V when the button is not pressed, so it can be said that the LED is “pulled up” to 5V, and will turn on. Leave room on your breadboard to display all constructed circuits at the same time!



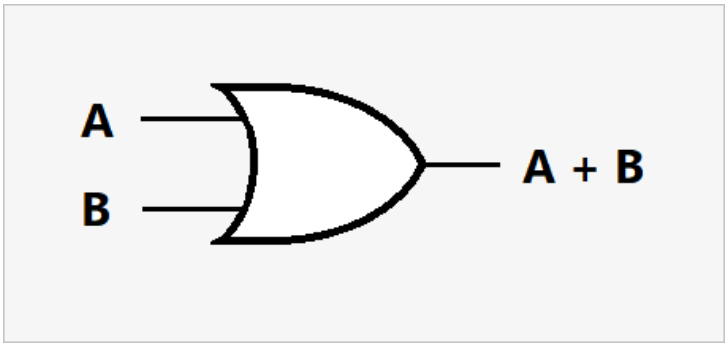
Pull-Down Resistor Schematic by Jean Braza

Next, construct the pull-down resistor circuit on your breadboard. This configuration has the opposite functionality of the pull-up resistor. When the pushbutton is pressed, the LED is connected to 5V, and turns on. When the pushbutton is not pressed, the LED is “pulled down” to GND, and turns off.

*Note: pull-up resistors are generally preferred in circuit design because many digital circuits use active-low as a convention. Here you can see that the pull-down resistor requires an additional resistor.

Part 2: Designing an OR circuit

In the second part of this project, we will apply our knowledge of pull-up and pull-down resistor configurations to both design and implement a simple circuit given its requirements.



An OR is a simple logic gate (more on digital logic in lecture 4!) whose output is true when either one of its inputs are true. Truth can be determined arbitrarily depending on whether we treat the output as active-high (true when 5V) or active-low (true when GND). Therefore, you are free to construct the circuit however you wish, using either pull-down or pull-up resistor configurations. Your final circuit should include 2 buttons, and an LED whose state toggles when either one or both of the LEDs is pressed.

*Hint: Start by drawing a circuit diagram first before implementing it on your breadboard. Designing a clear circuit diagram will make debugging easier.

***Bonus:** Challenge yourself to create an AND circuit! The AND gate is another fundamental digital logic component, and more information is readily available online.

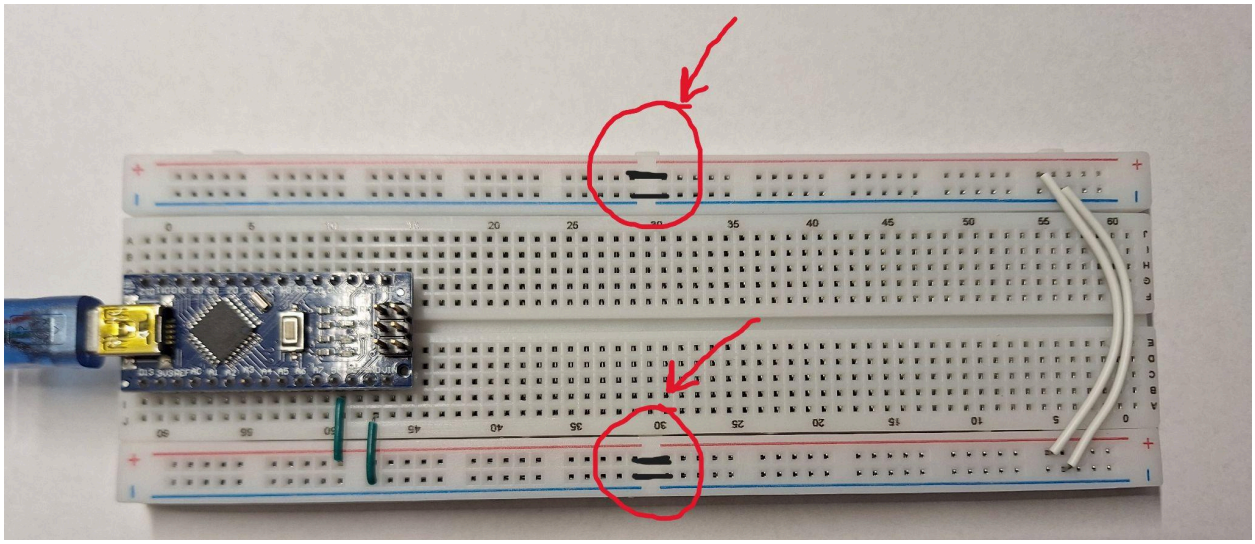
What to Submit

Due Date: Tuesday, 10/13/26

Submit your completed **Part 2** circuit demonstration to **UF Check-In** as an unlisted YouTube video. Remember, if you ever have trouble with OPS, we are here to help! Feel free to reach out on the Slack discussion channel, contact OPS coordinators directly, or attend office hours.

Resources

Lecture Slides
Hints:



*It is recommended that you connect both power rails to your Nano to power the entire breadboard. On our breadboards, the **power rails are not fully connected!** Small wires are necessary at the middle of the rails on either side if you want the entire rail to be powered.*

- Make sure that wires have a good connection to the holes in your breadboard by firmly pressing the exposed wire into the hole
- **Avoid connecting VCC directly to GND as this can damage parts!**

Sample Code: N/A
Schematics: N/A

Helpful Blogs:
Electronic Laws Overview
https://www.digikey.com/en/maker/tutorials/2024/electronics-laws-overview-theories-and-practical-use-cases
Water Analogy for Ohm's Law
https://www.digikey.com/en/maker/blogs/2020/electronics-beginning-with-the-basics